UNIVERSITY OF CAMBRIDGE

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The Spatial Distribution of Gravitational Wave Sources

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Abstract

Using data for 83 gravitational wave events from the latest LIGO observing run, we investigated the spatial distribution of gravitational wave sources. In particular, we looked for evidence of anisotropy in the population of binary black hole mergers. We performed a Bayesian model selection between an isotropic model and anisotropic models based on both `HEALPix` grids and spherical harmonics. We calculated Bayes factors to assess which model was preferred by the data and also computed posteriors for the population parameters of each model. We also investigated the importance of including selection effects. Our main result is a Bayes factor of 0.3 (Z_{aniso}/Z_{iso}), indicating no statistically significant evidence for anisotropy. As the number of active detectors increases, localisation of these sources will improve and the potential for reliable detection of anisotropy will increase. A subsequent analysis using a greater number of three-detector events will therefore be appropriate.

1 Introduction

As of April 2026, the LIGO-Virgo-KAGRA collaboration has officially detected 218 confident gravitational wave (GW) events across four observing runs since the first detection of a gravitational wave in 2015. LIGO operates two detectors in the United States, one in Livingston, Louisiana and one in Hanford, Washington. Virgo is located near Pisa, Italy and KAGRA is in Hida, Japan.

Following the success of LIGO, an international consortium is planning to develop a third-generation GW observatory known as the Einstein Telescope. Unlike the surface-level LIGO facilities in Hanford and Livingston, this European-led project will be constructed deep underground in order to shield it from seismic and human-induced noise. This subterranean design, coupled with other advanced technologies, will provide a significant leap in sensitivity. These improvements will allow for more precise source localisation and provide deeper insights into the populations of cosmic GW sources.

To date, no four-detector event has been observed. Three-detector events have been observed, but the majority still involve only LIGO's Livingston and Hanford interferometers. Additional detectors significantly improve the sky localisation of sources. This localisation is achieved via Bayesian parameter estimation using sampling methods such as MCMC or nested sampling. This inference has already been performed by the LIGO-Virgo-KAGRA collaboration, so we use their posterior sample estimates of the source parameters.

Our goal is to use this data, along with detector strain data (see Section 3.2 and Section 3.3 for more detail), to probe the anisotropy of the universe. We do this by considering how rates of GW events differ across the sky. This is an interesting goal since it tests the Cosmological Principle that the universe is homogeneous and isotropic on large scales. There are a few advantages of using GWs as opposed to EM radiation such as the CMB to investigate isotropy of the universe. Unlike EM radiation, GWs interact very weakly with matter and are not obscured by Galactic dust or the Earth's atmosphere. This takes away a layer of complexity that is present when using measurements of light.

Potential sources of true anisotropy in the spatial distribution of GW sources include galaxy clusters, the Earth's relative motion to the Cosmic Rest Frame and inflationary fluctuations.

Selection effects, discussed later, as well as nearby loud sources, introduce observed anisotropies, which we attempt to resolve in our analysis. Additionally, gravitational lensing by nearby galaxy clusters can introduce another observed anisotropy in the data. By magnifying signals from the distant universe, these clusters allow for the detection of sources that would otherwise fall below the sensitivity threshold. As a result, more events may be observed in the direction of these lenses, even if the true distribution of sources is uniform. In our analysis, we do not account for this particular selection effect. However, recent research incorporating these considerations find lensing-effects to have limited influence (Q. Yang et al [20]).

This research was heavily motivated by Stiskalek, Veitch, and Messenger (SVM) [1], who performed a similar analysis using posterior samples from 10 events in observing runs O1 and O2. They found a Bayes factor of $1.3:1$ in favour of isotropy, which was insufficient to pick one model over the other. Other articles such as Z. Li et al [2] and K. Gopal et al [3] also obtain results consistent with isotropy. Our subsequent analysis, which uses samples from an increased catalogue of 83 events from the most recent O4 parameter estimation release, is therefore warranted.

In our analysis, we adopt a hierarchical Bayesian framework following standard GW population inference (e.g. I. Mandel et al [4]). The population parameters describe the spatial distribution of GW sources. We perform three separate parameter inferences. Our main analysis includes selection effects and uses the Python Package `healpy` [6], described later, to specify the model. We then do two further analyses without considering selection effects. For one of these, the model is still specified using `healpy`. For the other, we use a basis of spherical harmonics on the sphere to describe the probability density of GW sources originating from certain

sky positions. For all three approaches, we compute posteriors for the corresponding population parameters. We also calculate Bayes factors to determine whether isotropy or anisotropy is preferred by the data.

2 Data

Throughout this project we use data published by LIGO after each observing run, downloaded from Zenodo as HDF5 files containing thousands of posterior samples for each source parameter. These parameters include the sky position angles, luminosity distance, and component masses. We use these samples in our statistical analysis and also for exploratory plots to verify that our code produces reasonable output.

Despite more data being available to us, for our analysis, we only use data from the 83 O4 events that were detected between 5 June 2023 and 9 January 2024. These are listed in Table 7. We omit the first 3 events from the beginning of the O4 parameter release since 2 of them are thought to have a neutron star–black hole binary source and the other doesn't include a set of Mixed posterior samples, which contains posterior samples from all of the waveform models. This leaves us with the 83 consecutive events with binary black hole sources that we use for our analysis.

As shown in Table 7, out of our 83 events, 7 of them are one-detector events. The rest are two-detector events.

The sensitivity of the detectors is ever increasing and so detection probabilities also increase significantly for later observing runs. Thus, as time passes, the detectors detect events at a higher rate. Since these probabilities are estimates, inferences on the magnitude of the total rate of GW events are likely to differ between runs. Including data from all 4 runs therefore risks introducing inconsistencies, potentially biasing the final result.

Additionally, since detection probabilities for earlier runs are smaller, the Monte Carlo estimation procedure, described in Section 3.3, yields exactly zero in grid cells when the true probability falls below the resolution set by the number of trials. In order to eliminate these zeros, we would need to use a much larger number of trials, which would increase the computational cost massively. This is another reason for only using the most recent events available to us.

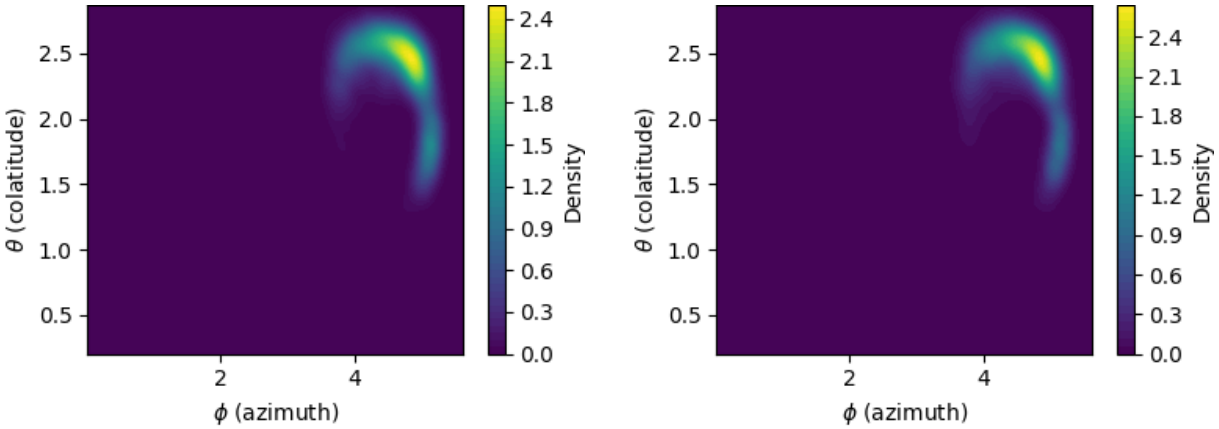
2.1 Differences Between Waveform Models

LIGO performs parameter inference using several different waveform models, each encoding the physics of a GW event by predicting the strain time series given a set of source parameters. Posterior samples computed using different waveform models are released, and we verified that different models produce similar sky-position posteriors.

We can use the sky position posterior samples from each waveform model to estimate the true posterior distribution. We did this using kernel density estimates. To quantify the distance between the two estimates of the posteriors, we used the Kullback–Leibler divergence:

$$D_{KL}(P \,\|\, Q) = \int_{-\infty}^{\infty} p(x) \log \frac{p(x)}{q(x)} \, dx \tag{1}$$

The KL divergence between the two distributions in Figure 1 consistently fell between 0 and 0.1 when we repeated this for many different events. These small differences are within the range of noise caused by the sampling process itself. We therefore concluded that any of the available waveform models would be sufficient for our purposes.



(a) IMRPhenomXPHM-SpinTaylor (b) SEOBNRv5PHM

Figure 1: Kernel density estimates for the sky-position posterior using samples from two different waveform models for the event GW230820. The KL divergence for this event and these two models was 0.05.

2.2 Detector Noise

Non-zero strain measured by the interferometers is predominantly caused by human activity and the environment as opposed to actual GW events. Factors such as passing trains and cars as well as stormy weather conditions can cause seismic ground motion, which can cause significant strain time series in the detector. This effect is referred to as detector noise.

The absolute values of the noise strains are orders of magnitude greater than signal strains. Thus, in order to detect signals, we need to look at the noise and signal strain curves in frequency space. Specifically, anthropogenic noise tends to manifest in a frequency range of $1-10\,\text{Hz}$. The Power Spectral Density (PSD) is a measure of the noise power of the detector as a function of frequency and is used to calculate signal to noise ratios as described in Section 3. Figure 2 illustrates the fact that the detector is significantly less noisy in the O4 run compared to the O1 run. Across the overwhelming majority of frequencies in the range plotted above, the power of the noise for O1 is greater than for O4. In some cases, the difference is several orders of magnitude but, in most cases, there is roughly an order of magnitude difference. This contributes to significantly increased rates of event observations and the ability to detect sources from much further distances.

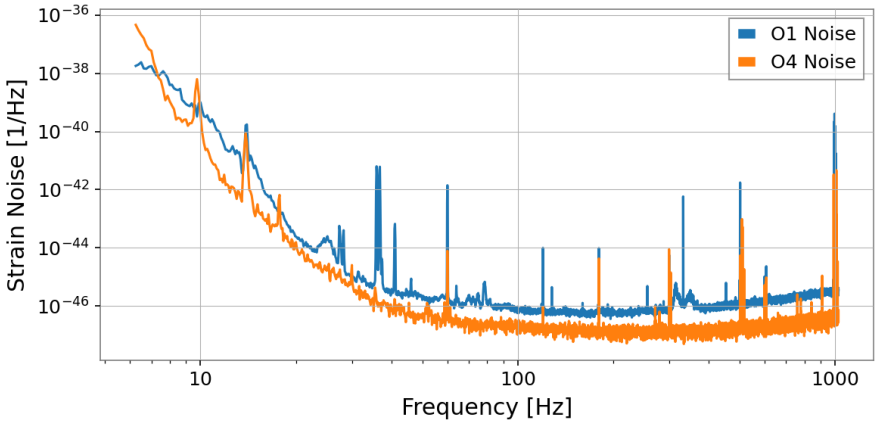


Figure 2: Two PSDs for the Hanford detector plotted against frequency. The one in blue is from the O1 observing run . The one in orange is from the most recent O4 observing run.

3 Method

3.1 Initial Exploration with `healpy` and `HEALPix`

The sky is divided into equal-area pixels using the `HEALPix` scheme, which we implement via the `healpy` Python package. This framework provides the basis for our population inference and allows us to generate Mollweide projections and scatter plots of various sky maps. Figure 3 and Figure 4 were produced early in the project to gain hands-on experience with `healpy` and to check for any obvious anisotropy.

In Figure 3, the different sets of samples form roughly circular arcs, which is typical for two-detector events. This is due to the time delay between the wave being detected at Hanford and Livingston. Maximal time delay is achieved by waves travelling in the direction along the ray connecting the two detectors. The straight-line distance between the interferometers is roughly 3000 km and GWs propagate at the speed of light. This maximal delay is therefore approximately 10 ms. Waves detected at the same time by the two detectors travel in the plane bisecting the two detectors. These directions form a great circle on the sphere.

The binned posterior samples (Figure 4) show no striking anisotropy; however, the four pixels along the equatorial plane exhibit slightly lower frequencies compared to those in the northern and southern regions. This observation is somewhat expected since Figure 6 shows that the detectors are, on average, less sensitive to waves around the equatorial plane. Note that here we use the default HEALPix pixel structure.

3.2 Selection Effects

Gravitational wave strain measures the difference in the respective lengths of the two arms of an interferometer relative to their original length, which is induced by the passage of a wave. It is defined by the expression $h = \frac{\Delta L}{L}$ where ΔL is the difference in length and L is the original length.

Another expression for the strain of a particular detector in terms of the metric perturbation is:

$$h = h_+ F_+ + h_\times F_\times \tag{2}$$

h_+ and $h_\times$ are known as the "plus" and "cross" polarisation amplitudes respectively. They are the components of the metric perturbation in Transverse-Traceless gauge. F_+ and $F_\times$ are known as the antenna response functions. These are completely dependent on the relative geometry between the interferometer arms and the polarisation of the GW source. This illustrates why detectors have different sensitivities to identical sources located at different points in the sky.

Figure 5, showing how the probability of detecting a GW varies across the sky, further illustrates that the interferometers have a sky-position-dependent sensitivity. The two clear antipodal peaks correspond to the theoretical sensitivity maxima directly above and below the plane of the detector arms. Hanford and Livingston are close together on the scale of the Earth. The two respective planes of the detector arms are separated by an angle of 27.2°. As a result, these peaks appear approximately above and below the United States. Unlike in Figure 5, for the actual analysis, we use a distance prior extending to 5 Gpc. The spatial structure of the antenna pattern is unchanged, but the absolute values of the probabilities differ.

We also observe that the posterior samples tend to coincide with regions of high detection probability. This reflects the fact that LIGO's posterior on sky position naturally biases towards such directions, particularly for two-detector events where the time-delay localisation alone is poor. For three-detector events this trend is less pronounced.

Figure 6 shows that the time-averaged detection probability does not vary dramatically with sky position. There are modest minima around the equatorial plane, meaning the detector observes a lower proportion of the GWs propagating from these directions. Assuming a uniform

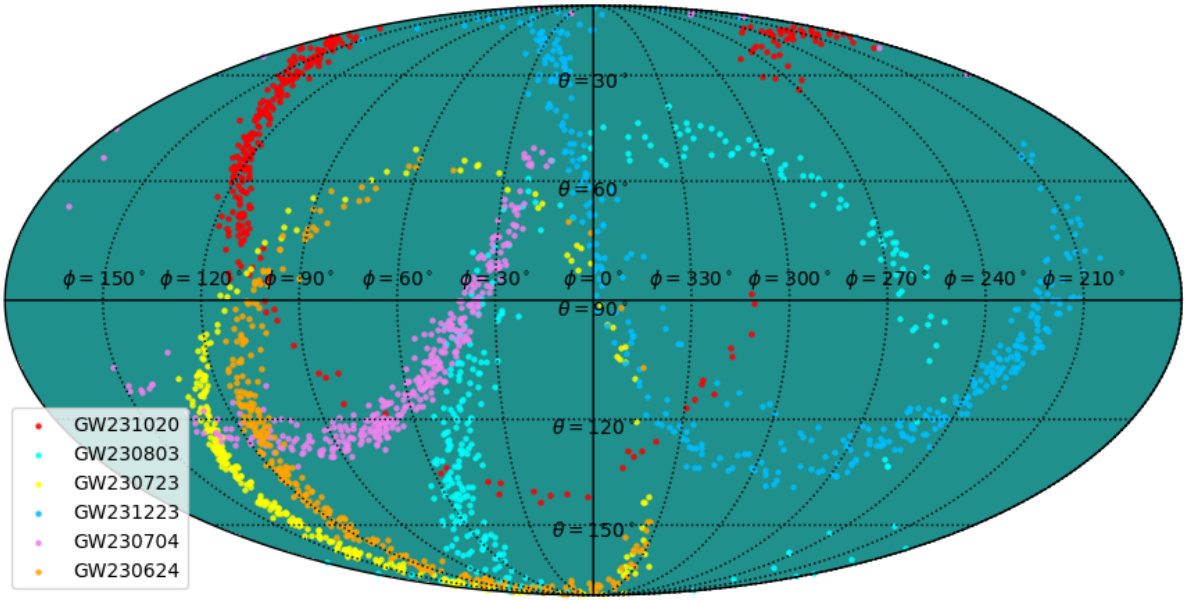


Figure 3: Posterior samples from 6 arbitrarily chosen events scattered across a Mollweide projection of the sky. We label lines of constant θ and constant ϕ in degrees. These are spherical polar coordinates for the celestial sphere related to ra and dec via $\theta = 90° - dec$ and $\phi = ra$. This is the convention we will use for all subsequent Mollweide projections.

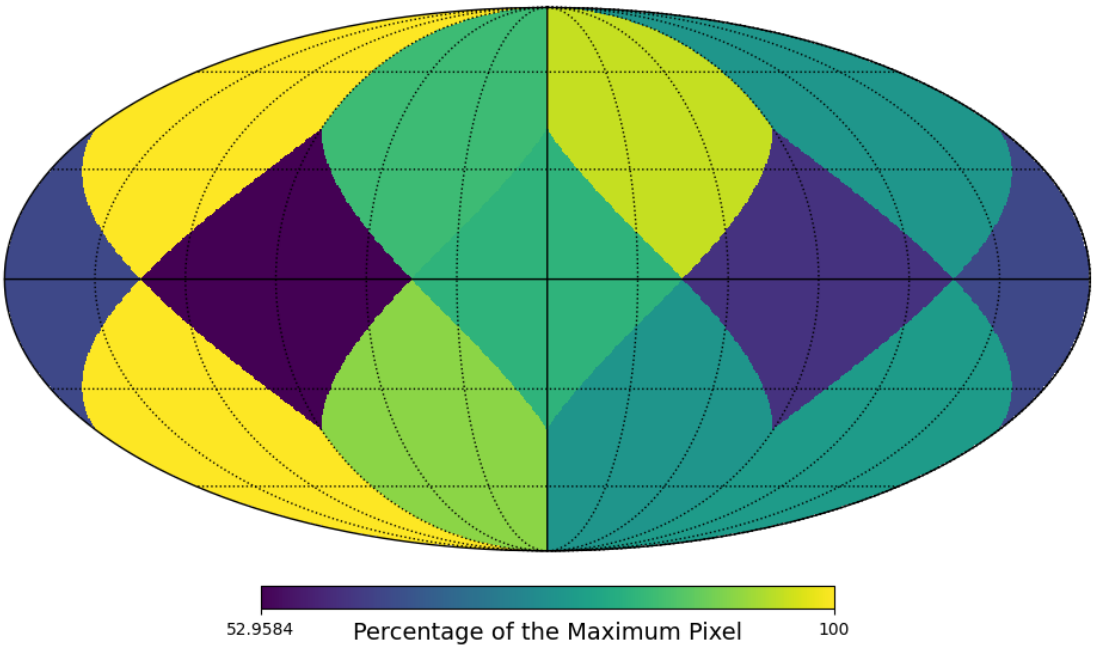
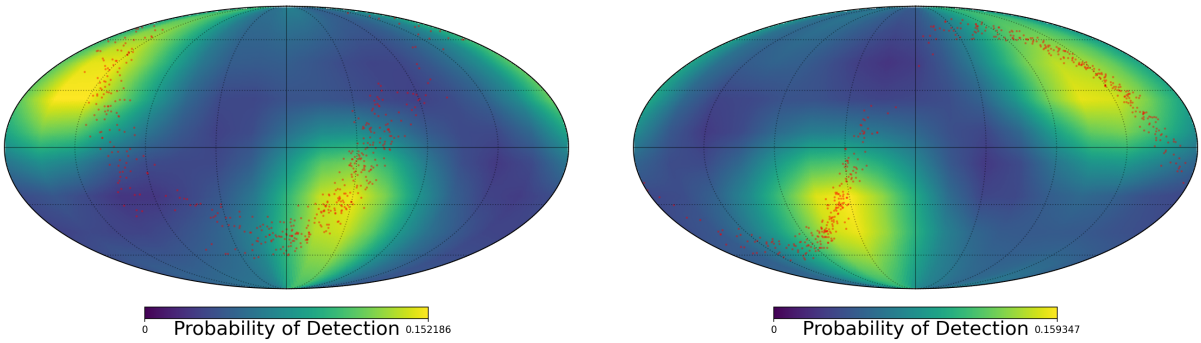


Figure 4: A HEALPix grid with 5000 posterior samples from each event binned into 12 pixels, including events from O1, O2, O3 and O4. Note that this figure includes events from earlier observing runs that are not listed in Table 6 and Table 7. The colour map shows the percentage of samples in each pixel relative to the pixel with the most samples.



(a) At the time of GW230704

(b) At the time of GW230928

Figure 5: Detection probability $p(\text{det} \mid T, \Omega)$ as a function of sky position at the times of two detected events, GW230704 and GW230928. Overlaid in red are posterior samples for each event. A distance prior of 3 Gpc was used for visual clarity.

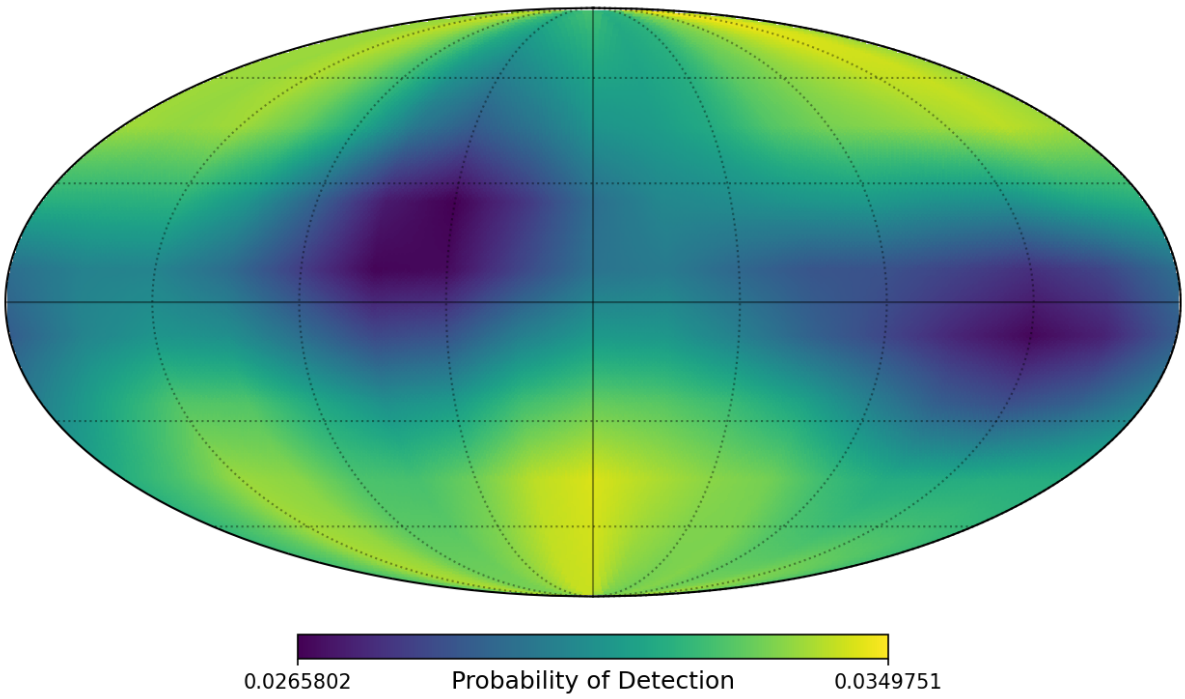


Figure 6: Time-averaged detection probability across the O4 observing run.

rate of events in all directions, we would therefore expect to detect fewer events from these angles.

Another important selection effect is luminosity distance. Figure 7 shows the marginalised detection probability as a function of distance for 6 different GW events from the O4 observing run. The probability of detecting an event at 5 Gpc away is small, but our distance prior , $p(d_L) \propto d_L^2$ (consistent with a flat Euclidean universe), means source density increases with distance. During O4, events with inferred luminosity distances up to 7.5 Gpc were detected. The ability to detect events at such large scales may warrant use of a more sophisticated cosmological model that includes non-zero curvature at large redshifts. Such changes to the model may necessitate use of a different distance prior as the volume calculation depends on the metric and therefore the curvature. Our analysis, however, assumes a flat universe.

3.3 Computing Detection Probabilities

In order to estimate detection probabilities at fixed time and sky position we perform a Monte Carlo sum to average over the rest of the source parameters. We use a set of priors to randomly draw the rest of the relevant source parameters. Given a full set of source parameters, we generate a waveform using the Python package `Bilby` [12]. For a detector of choice, this gives us a theoretical prediction for the strain time series produced by such a source. With this and the PSD for the interferometer at the time, we can calculate the optimal signal to noise ratio for the source. The optimal SNR is given by:

$$\rho_i(\theta) = 4 \int \frac{|h_i(f,\theta)|^2}{S_i(f)} df \tag{3}$$

where $h_i(f,\theta)$ is the Fourier transform of the strain time series, $S_i(f)$ is the power spectral density (PSD) and i indexes the interferometers. The strain and therefore the SNR depends explicitly on the source parameters denoted by θ. Then, if there are I interferometers active, the network SNR is:

$$\rho(\theta) = \sqrt{\sum_{i=1}^{I} \rho_i(\theta)^2} \tag{4}$$

If ρ exceeds some threshold SNR, ρ_{thresh}, then the event is detected. In order to estimate the detection probabilities, we repeatedly draw full sets of source parameters from our priors. The detection probability is then the fraction of the sets with $\rho > \rho_{thresh}$.

In the analysis of Subsections 3.4, 3.5, 3.6, 3.7, and 3.8, the detection probabilities are often conditional on the time the event was observed. Thus, the correct way to compute detection probabilities is to use the PSDs of the detectors at these times. However, using per-event PSDs for a large number of events would have introduced additional complexity without adding much to the inference. This approximation does introduce some bias to the absolute values of any inferred merger rates but is unlikely to affect conclusions regarding anisotropy.

It was not feasible to compute detection probabilities and generate waveforms within our likelihood function code. It meant that the run time for one evaluation of the likelihood function (Subsection 3.6) was far too large for our dynamic nested sampling algorithm (Subsection 3.11) to run in a reasonable time. To mitigate this, we precomputed a grid of detection probability values at fixed time and fixed sky position. For each of the 83 detection times of events in our catalogue, we evaluate the detection probability for each entry of an 8 by 16 grid of sky positions. This grid consists of 8 θ and 16 ϕ values. Here, θ and ϕ are spherical polar coordinates as defined in Figure 3. Then, in order to approximate detection probabilities at positions not in the grid, we use linear interpolation between grid points on the sphere. Since we use a large number of grid points, this approximation is valid.

We calculated the detection probabilities assuming that all 83 events were two detector events. As a result, the detection probabilities we calculate for the 7 one detector events will

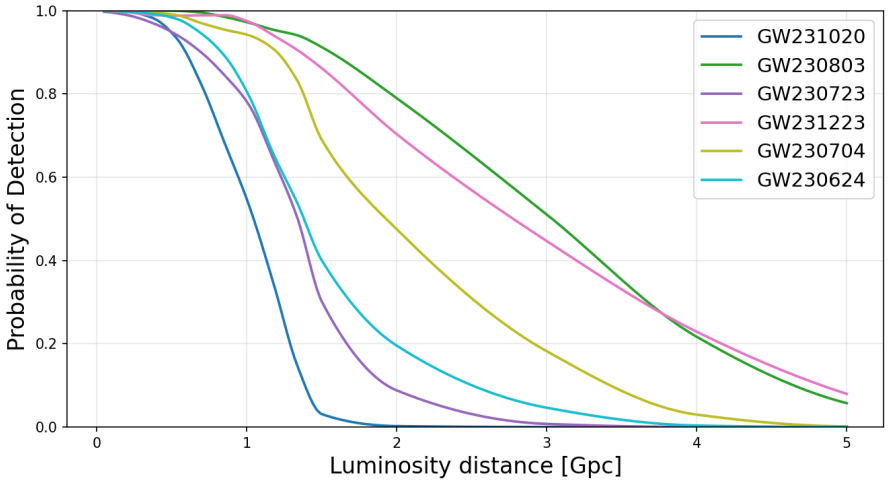


Figure 7: Detection probability marginalised over source parameters other than luminosity distance for the 6 GW events plotted in Figure 3.

be marginally wrong. However, the antenna patterns, as shown in Figure 5, will be exactly the same. This almost certainly will not affect our conclusions.

3.4 Inference

To conduct our inference, we consider two different models, an isotropic and an anisotropic model. Each model is specified by its population parameters, which describe the true distribution of GW events across the sky. To determine which model is preferred by the data, we perform a Bayesian model comparison.

The anisotropic model is the more complex of the two, consisting of 15 population parameters. We use the Python package `healpy` to split the sky into 12 equally sized pixel regions. Of our 15 parameters, 12 are rate parameters $(a_1, \ldots, a_{12})$ describing the rate of GW events in each pixel. These pixel weights are the number of GW events per unit volume per unit time per unit steradian. We use units of $[\mathrm{Gpc}^{-3}\,\mathrm{sr}^{-1}\,\mathrm{yr}^{-1}]$ to express these pixel weights throughout the report. The overall rate is related to the pixel weights by:

$$R = \Delta\Omega \sum_{i=1}^{N_{\mathrm{pix}}} a_i \tag{5}$$

where $\Delta\Omega = \frac{4\pi}{N_{\mathrm{pix}}}$ is the solid angle taken up by each pixel.

The remaining three are Euler angles, (α, β, γ), parametrising a rotation of the HEALPix grid. We use the ZXZ Euler angle convention [5]. This rotational freedom ensures we do not miss anisotropy as a result of a fixed pixel structure.

The isotropic model assigns a fixed rate to all sky regions and therefore requires only one population parameter, the total rate R, which is the number of GW events per unit volume per unit time. We express R in units of $\mathrm{Gpc}^{-3}\,\mathrm{yr}^{-1}$ throughout this report. Since the rate is the same across the sky, rotations of the pixel grid are redundant. Effectively, the isotropic model is just the anisotropic model where we restrict the population parameter space to parameters with equal pixel weights. The only thing we change is the prior. The likelihood function, explored in Section 3.6, is the same but is restricted to a smaller set of arguments.

Throughout the remainder of this report, we denote the full vector of population parameters for the anisotropic model as λ:

$$\lambda = (a_1, ..., a_{12}, \alpha, \beta, \gamma) \tag{6}$$

3.5 Evidence

To perform a Bayesian model comparison, we compute the evidences for each of the models. The evidence is our likelihood function averaged over all population parameters. This is given by:

$$p(N_{obs}, \{d_j\} \mid det) = \int p(N_{obs}, \{d_j\} \mid det, \lambda)\, p(\lambda)\, d\lambda \tag{7}$$

where λ are our population parameters, *det* indicates that we are conditioning on the event being detected and N_{obs} is the number of events observed in the observing run. $\{d_j\}$ refers to the data used by LIGO to generate posterior samples on the source parameters of event j. This of course includes but is not restricted to the strain time-series data in each of the detectors around the time of event j and the time of detection at both Hanford and Livingston. $p(\lambda)$ is the prior probability density function on the population parameters.

Of course the evidence changes depending on which model we are computing it for. For the anisotropic model, the right hand side of Equation 7 is a 15 dimensional integral. For the isotropic model, it is a 1 dimensional integral.

3.6 Likelihood Function

The full likelihood factorises as:

$$p(\{d_j\}, N_{\rm obs} \mid \det, \lambda) = p(N_{\rm obs} \mid \det, \lambda)\, p(\{d_j\} \mid \det, \lambda) \tag{8}$$

where $p(N_{\rm obs} \mid \det, \lambda)$ is the "number likelihood" and $p(\{d_j\} \mid \det, \lambda)$ is the "event likelihood". As we will show, the event likelihood depends only on the relative sizes of the pixel weights, while the number likelihood controls the absolute scale and takes into account the rate at which events are observed.

3.7 Event Likelihood

Given $N_{\rm obs}$ events with separate datasets $\{d_j\}$, the event likelihood factorises as:

$$p(\{d_j\} \mid \det, \lambda) = \prod_{j=1}^{N_{\rm obs}} p(d_j \mid \det, \lambda) \tag{9}$$

Each factor, $p(d_j \mid \det, \lambda)$, depends implicitly on the detector geometry and noise curves at the time of detection, which we make explicit by writing $p(d_j \mid \det, T_j, \lambda)$ where T_j is the detection time.

Applying Bayes' theorem and marginalising over source parameters θ:

$$\begin{aligned} p(d_j \mid \det, T_j, \lambda) &= \frac{p(\det \mid d_j, T_j, \lambda)\, p(d_j \mid T_j, \lambda)}{p(\det \mid T_j, \lambda)} \\ &= \frac{\int d\theta\, p(d_j \mid T_j, \theta)\, p(\theta \mid T_j, \lambda)}{\int d\theta\, p(\det \mid T_j, \theta)\, p(\theta \mid T_j, \lambda)} \\ &= \frac{\int d\theta\, \frac{p(\theta \mid T_j, d_j)\, p(d_j \mid T_j)}{\pi(\theta)}\, p(\theta \mid T_j, \lambda)}{\int d\theta\, p(\det \mid T_j, \theta)\, p(\theta \mid T_j, \lambda)} \end{aligned} \tag{10}$$

where θ refers to the two angles specifying the sky position of the source and π is the prior on the sky position used by LIGO for their inference of the posterior samples. We have also used the fact that $p(\det \mid d_j, T_j, \lambda) = 1$ since event j is in our catalogue and has already been detected at time T_j. Once we know the PSDs for both detectors (encoded by the conditioning on T_j) and the strain time-series of both detectors (encoded by the conditioning on d_j), whether the event is detected is deterministic.

The numerator of Eq. (10) can be approximated using a Monte Carlo sum over posterior samples:

$$\int d\theta \, \frac{p(\theta \mid d_j)\, p(d_j)}{\pi(\theta)} \, p(\theta \mid \lambda) \approx \frac{p(d_j)}{S_j} \sum_{i=1}^{S_j} \frac{p(\theta_j^i \mid \lambda)}{\pi(\theta_j^i)} \tag{11}$$

where S_j is the number of samples used and θ_j^i is the ith sample for the jth event. Since $p(\theta_j^i \mid \lambda)$ is fully controlled by the pixel weights:

$$\frac{p(d_j)}{S_j} \sum_{i=1}^{S_j} \frac{p(\theta_j^i \mid \lambda)}{\pi(\theta_j^i)} \propto \sum_{i=1}^{S_j} a_j^i \tag{12}$$

where a_j^i is the pixel weight corresponding to the pixel containing the ith sample under the grid rotation specified by the Euler angles in λ. In practice, we want to work out which pixel a certain sample is in given some set of Euler angles. In order to do this, we apply the inverse rotation to the sample and check which pixel from the default grid the result lies in. This is completely equivalent to fixing the sample positions, rotating the grid by the actual rotation and checking which pixel the sample lies in.

The denominator can be approximated as:

$$\int d\theta \, p(\text{det} \mid T_j, \theta)\, p(\theta \mid T_j, \lambda) \propto \Delta\Omega \sum_{i=1}^{N_{\text{pix}}} a_i \, p(\text{det} \mid T_j, \Omega_i) \tag{13}$$

where Ω_i are the rotated pixel centres. This is an approximation because the detection probability varies throughout each pixel region, but we evaluate it only at the centre.

Then the full expression for the likelihood of one event is given by:

$$p(d_j \mid \text{det}, T_j, \lambda) \propto \frac{\sum_{i=1}^{S_j} a_j^i}{\Delta\Omega \sum_{i=1}^{N_{\text{pix}}} a_i \, p(\text{det} \mid T_j, \Omega_i)} \tag{14}$$

This demonstrates that the event likelihood is invariant under a scaling of the pixel weights and therefore depends only on their relative values. The numerator is large when the pixel weights corresponding to pixels containing posterior samples are large. The denominator is small when the pixel centres corresponding to high rate pixels are located at sky positions with low detection probabilities, $p(\text{det} \mid T_j, \Omega_i)$. So the overall fraction is large when high pixel weights are assigned to the pixels containing lots of posterior samples and these pixels are in regions of the sky with low detection probabilities.

The likelihood is therefore preferentially upweighted when posterior samples lie in regions of low detection probability. This makes sense since observing events at sky locations where the detector is least sensitive provides the strongest evidence for genuine source anisotropy.

3.8 Number Likelihood

The number likelihood is not simply the probability of observing a given count of events; it is the probability of observing N_{obs} events at the specific times they occurred.

We split the observing run into n intervals of length T/n each, with n chosen to be large enough that the probability of detecting more than one event per interval is negligible. The number likelihood becomes:

$$p(N_{\text{obs}} \mid \lambda) \propto \prod_{i \in S} p(N \geq 1 \mid \lambda, T_i) \times \prod_{j \in S'} p(N = 0 \mid \lambda, T_j) \tag{15}$$

where S and S' are the sets of intervals with and without detections respectively.

Modelling detections as a Poisson process, for each interval:

$$p(N = 0 \mid \lambda, T_i) = \exp(-\hat{N}_i(\lambda)) \tag{16}$$

$$p(N \geq 1 \mid \lambda, T_i) \approx p(N = 1 \mid \lambda, T_i) \tag{17}$$

$$= \hat{N}_i(\lambda) \exp(-\hat{N}_i(\lambda)) \tag{18}$$

where $\hat{N}_i(\lambda)$ is the expected number of detections in interval i.

Substituting into Equation (16):

$$p(N_{\text{obs}} \mid \lambda, \text{det}) \propto \prod_{i \in S} \hat{N}_i \exp(-\hat{N}_i) \times \prod_{j \in S'} \exp(-\hat{N}_j)$$

$$= \prod_{i \in S} \hat{N}_i \times \exp\left(-\sum_{j=1}^{n} \hat{N}_j\right) \tag{19}$$

The expected number of events in each interval is:

$$\hat{N}_i(\lambda) = \frac{T}{n} V_L \int d\Omega \, \frac{dR}{d\Omega} \, p(\text{det} \mid T_i, \Omega) \tag{20}$$

where $dR/d\Omega$ is the differential rate and V_L is the prior volume.

By definition:

$$R = \Delta\Omega \sum_{i=1}^{N_{\text{pix}}} a_i = \int \frac{dR}{d\Omega} \, d\Omega \tag{21}$$

so within pixel j the differential rate equals a_j. Equation (20) then becomes:

$$\hat{N}_i = \frac{T}{n} V_L \, \Delta\Omega \sum_{j=1}^{N_{\text{pix}}} a_j \, p(\text{det} \mid T_i, \Omega_j) \tag{22}$$

where Ω_j is the rotated central angle of pixel j. As in Equation 13, this is an approximation since $p(\text{det} \mid T_i, \Omega)$ may vary throughout each pixel. This approximation can be improved by using more pixels to partition the integral.

Finally, the argument of the exponential in Equation (19) is:

$$\hat{N} = \sum_{i=1}^{n} \hat{N}_i$$

$$= T \, V_L \, \Delta\Omega \sum_{j=1}^{N_{\text{pix}}} a_j \, \langle p(\text{det} \mid \Omega_j) \rangle \tag{23}$$

where $\langle p(\text{det} \mid \Omega_j) \rangle = \frac{1}{n} \sum_{i=1}^{n} p(\text{det} \mid T_i, \Omega_j)$ is the time-averaged detection probability at angle Ω_j over the course of the observing run. Figure 6 shows this quantity on a Mollweide projection. $\hat{N}$ is the expected number of detections over the whole observing run.

Because $\hat{N}$ appears inside an exponential, multiplicative constants cannot be ignored, so physically accurate values of T and V_L are essential.

3.9 Priors on Source Parameters

In order to calculate detection probabilities, as described in Section 3.3, we marginalise over all relevant source parameters other than time, right ascension and declination. In order to do this in practice, we need to be able to sample different sets of source parameters in a way that is physically reasonable. Below we list which priors were used.

Parameter Name	Prior Range	Prior Distribution
Mass 1 (Primary)	$[10\,M_\odot, 150\,M_\odot]$	$\propto m_1^{-2.35}$
Mass 2 (Secondary)	$[10M_\odot, M_1]$	Uniform
Luminosity Distance	$[100\,\mathrm{Mpc}, 5000\,\mathrm{Mpc}]$	$\propto {d_L}^2$
Inclination	$[-\frac{\pi}{2}, \frac{\pi}{2}]$	Cosine
Polarisation	$[0, \pi]$	Uniform
Coalescence Phase	$[0, 2\pi]$	Uniform

Table 1: Table shows the range and form of the distribution for the source parameters.

We use a Salpeter prior [7] for the largest mass in the binary system. To date, the largest inferred primary mass for a GW source is around $137\,M_\odot$. A prior bounding the primary mass by $150\,M_\odot$ was therefore chosen. The secondary mass is then sampled uniformly from the interval bounded by $10\,M_\odot$ and the primary mass.

For luminosity distance, we adopt a prior $p(d_L) \propto d_L^2$ over the range $[0, 5]\,\mathrm{Gpc}$, consistent with a uniform distribution of sources in a flat Euclidean universe.

The upper bound of 5 Gpc covers the bulk of the posterior samples across our O4 catalogue. A small number of events have samples exceeding this bound with a maximum sample for luminosity distance at around 7.5 Gpc. These tails are mildly truncated by our prior choice. However, this is not expected to affect our inference significantly. The decrease in the average value of detection probabilities caused by increasing this distance upper bound is compensated by the increase in V_L. This ensures that the expected number of detections in Equation 20 is unchanged. We tested this by reducing our upper bound from 5 Gpc to 3 Gpc and saw that the size of the inferred pixel weights was roughly unchanged. On the other hand, the inferred total rate is sensitive to the mass prior we choose. Increasing the upper bound of the primary mass prior from $50\,M_\odot$ to $150\,M_\odot$, resulted in the inferred total rate roughly halving.

The rest of the priors are standard priors used in GW astronomy for inclination, polarisation and coalescence phase.

3.10 Priors on Population Parameters

For the Euler angles, we use uniform priors with $\alpha \in [0, \frac{\pi}{4}]$, $\beta \in [0, \frac{\pi}{2}]$ and $\gamma \in [0, \frac{\pi}{4}]$. We do this to limit the number of sets of Euler angles that simply permute the `healpy` pixels. Given fixed pixel weights, such sets of angles result in equal likelihoods and will therefore produce multimodal posterior distributions.

For both models, we insist that $R \in [1, 1000]\,\mathrm{Gpc}^{-3}\,\mathrm{yr}^{-1}$. We choose a rate prior of $p(R) \propto R^{-\frac{1}{2}}$, which is a Jeffreys prior for a Poisson process. For the isotropic model, this is sufficient. For the anisotropic model, we draw R from the above distribution and then draw the pixel weights uniformly on the simplex, $\Delta\Omega \sum_{i=1}^{N_{pix}} a_i = R$. We sample $\{b_1, ..., b_{12}\}$ from a Dirichlet$(1, 1, ..., 1)$ distribution [8] where $a_i = \frac{R}{\Delta\Omega}\, b_i$ and $\sum_{i=1}^{N_{pix}} b_i = 1$. This is a suitable prior since it is symmetric under exchange of two different pixel weights and it makes no a priori assumption about the level of anisotropy.

3.11 Sampling

In order to calculate the Bayes factor and posteriors for population parameters, we use the Python package `dynesty` [13]. Given a likelihood function and a prior on the population parameter space, this uses a dynamic nested sampling algorithm to generate posterior samples and compute evidences to a certain level of uncertainty. It also allows us to resample from the resulting posteriors and produce visual representations of the results such as corner plots. For all of our results, we sample using 500 live points.

3.12 Pipeline Validation

Throughout the project we performed sanity checks to ensure the pipeline produced physically reasonable results. This subsection presents two of them.

To check the number likelihood, we computed it for a range of total rates under a uniform sky distribution (all pixel weights equal, zero Euler angles).

Under our 5 Gpc distance prior, the time-averaged detection probability across the sky is approximately $\langle p_{\rm det} \rangle \approx 0.035$ (Figure 6). The prior volume is $V_L \approx 524\,{\rm Gpc}^3$ and the observing interval is $T \approx 0.597$ yr. Using these quantities, we estimate the total rate via:

$$R_{\rm est} = \frac{N_{\rm obs}}{V_L\, T\, \langle p_{\rm det} \rangle} \tag{24}$$

$$\approx \frac{83}{524 \times 0.597 \times 0.035} \approx 8.5\,{\rm Gpc}^{-3}\,{\rm yr}^{-1} \tag{25}$$

The agreement between our back-of-the-envelope estimate and the numerical maximum in Figure 8 confirms that the number likelihood is behaving as expected. At this peak value of R, with equal pixel weights, the expected number of total detections across our observing window is 82.61. This is reassuring since the overall number of events in our time window is 83.

We also verified our code for SNR calculation by computing optimal SNRs for events with known published values. For each event we drew sets of source parameters from the posterior samples, used the PSD for that event, and computed the optimal SNR.

Event	5th pct.	95th pct.	Mean	Published
GW150914	19.16	26.59	23.34	26
GW151226	11.33	15.12	13.27	13
GW190412	18.09	21.39	19.76	19
GW170818	10.31	16.15	12.95	11
GW231020	9.16	13.52	11.38	12

Table 2: Optimal SNR validation against published values. For each event we show the 5th and 95th percentile and mean optimal SNRs computed from 100 posterior samples, compared to the published matched-filter SNR.

Table 2 shows agreement between our optimal SNRs and LIGO's published matched-filter SNRs across events spanning SNRs from 11 to 24. Full agreement is not expected because optimal SNR and matched-filter SNR are different quantities, and we use only 100 posterior samples which may not include the parameters yielding the maximum SNR. What matters for our pipeline is that the function correctly predicts whether sources with given parameters would be detected; Table 2 demonstrates that it does.

4 Results

In the following section, we present the results of our model comparison for the full `HEALPix` model including selection effects, the HEALPix model without selection effects and the model that uses spherical harmonics without selection effects.

4.1 Full Model Results

Figure 9 shows that Pixel 0 has the highest median pixel weight at $1.26\,{\rm Gpc}^{-3}\,{\rm sr}^{-1}\,{\rm yr}^{-1}$ whilst Pixel 6 has the smallest median at $0.20\,{\rm Gpc}^{-3}\,{\rm sr}^{-1}\,{\rm yr}^{-1}$, which is roughly 6 times less than that of Pixel 0. Pixels 0, 3, 8 and 10 clearly have the four largest medians. The other 8 pixels have medians within a $0.38\,{\rm Gpc}^{-3}\,{\rm sr}^{-1}\,{\rm yr}^{-1}$ range.

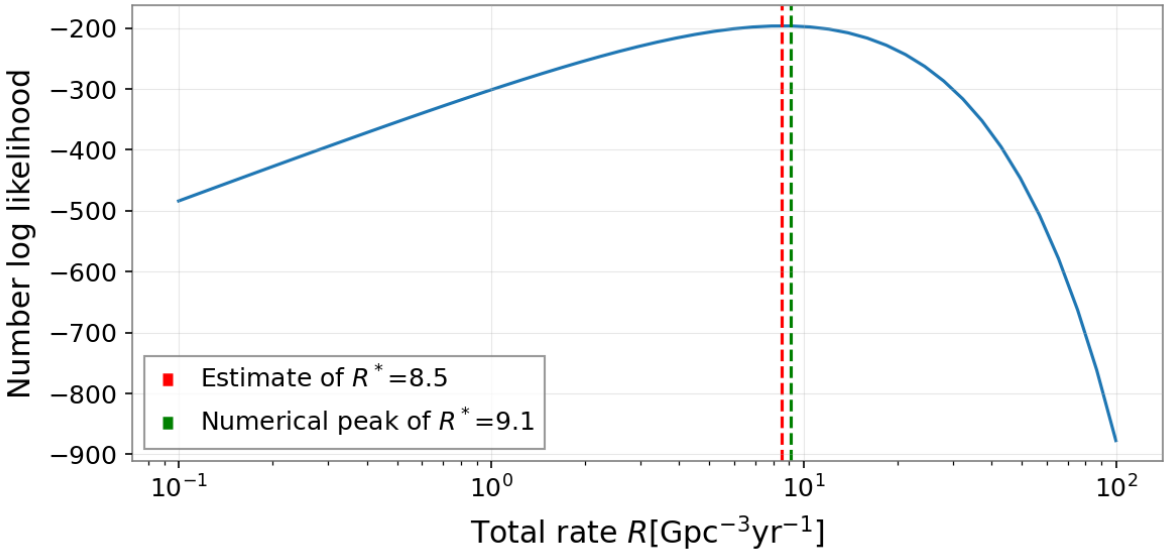
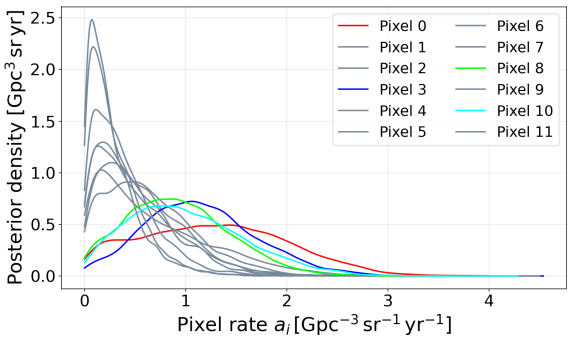
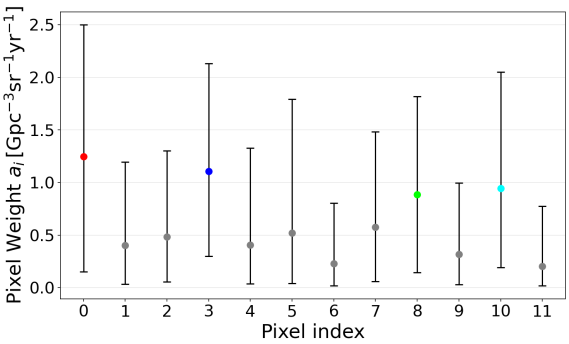


Figure 8: Number log-likelihood as a function of total rate assuming uniform pixel weights. The green dashed line marks the numerical maximum of the number likelihood; the red line marks our back-of-the-envelope prediction.



(a) Posterior marginal distributions for each of the 12 pixel weights. The distributions of the pixels with the highest medians are in colour.



(b) Median of the posteriors for each of the 12 pixel weights along with 5th to 95th credible intervals for each pixel.

Figure 9: Above plots represent the marginal posterior distributions for each of the pixel weights.

Figure 9a represents similar information to Figure 9b. However, we can see that these four highest rate pixels have maximum posterior density at values away from zero. They also have significant probability of taking values greater than $2\,\mathrm{Gpc}^{-3}\,\mathrm{sr}^{-1}\,\mathrm{yr}^{-1}$. On the other hand, the 8 other distributions have much sharper maxima for the weights close to 0.

The three 1D marginals in Figure 10 show that there is significant probability density spread across the prior ranges for each of the 3 Euler angles. This indicates that the inference has been unable to find specific rotations that fit the data well.

From Table 3, we can see that the estimates for the total rate, R, are similar in both models with their medians differing by roughly $0.06\,\mathrm{Gpc}^{-3}\,\mathrm{yr}^{-1}$. The isotropic model slightly favours lower total rates. These inferred rates fall slightly below the 90% credible interval of $17.9 - 44\,\mathrm{Gpc}^{-3}\,\mathrm{yr}^{-1}$ (R. Abbott et al [21]) found using data from the third observing run. However, this can easily be explained by differences in the mass prior and also various approximations made throughout our analysis.

Table 3 also shows that we compute a Bayes factor of 0.3. This indicates that our data fits the isotropic model marginally better than the anisotropic model. According to Jeffreys' scale, whilst close to being classed as substantial evidence, this Bayes factor is inconclusive.

Figure 11 shows the posterior distributions for R in both models. This is generated by sampling population parameters from each of the two posteriors, calculating the corresponding values for R, then binning these values into different equally sized bins. The two distributions are extremely similar, which is to be expected. We can see that the anisotropic model has a slightly larger median and both models have a very similar variance for R. This matches the results in Table 3.

Knowing about inferred rates for specific pixels doesn't tell you about rates at precise points in the sky. This is because the Euler angle parameters rotate the pixels, which means the pixel weight of that pixel may actually correspond to a slightly different region of the sky. Of course, we restrict the prior ranges on the angles, reducing this effect. However, Figure 12 gives us a clearer idea of how the rate, on average, differs across the sky. In Figure 12, we can see that the maximum inferred rate of approximately $1.44\,\mathrm{Gpc}^{-3}\,\mathrm{sr}^{-1}\,\mathrm{yr}^{-1}$ occurs close to the north pole. This is between 6 and 7 times the minimum inferred rate of roughly $0.23\,\mathrm{Gpc}^{-3}\,\mathrm{sr}^{-1}\,\mathrm{yr}^{-1}$. As expected this is similar to what we find when we compare the maximum and minimum median pixel weights.

In order to generate Figure 12, we used the posterior samples produced by the nested sampling. To produce the plot, we make use of a separate higher resolution pixel grid with 49152 pixels. Each posterior sample consists of 12 rates and 3 Euler angles. For each of the 49152 fixed pixel centres, we perform the inverse rotation to work out which new rotated pixel from the 12 pixel grid that centre lies in. We then assign the rate corresponding to that pixel from the 12 pixel grid to the pixel from the higher resolution grid. We do this for many posterior samples and take an average for each of the 49152 pixel centres.

Anisotropic total rate R_{aniso}	$8.60^{+1.64}_{-1.46}\,\mathrm{Gpc}^{-3}\,\mathrm{yr}^{-1}$
Isotropic total rate R_{iso}	$8.54^{+1.65}_{-1.41}\,\mathrm{Gpc}^{-3}\,\mathrm{yr}^{-1}$
$\log Z_{\mathrm{aniso}}$	-118.8 ± 0.2
$\log Z_{\mathrm{iso}}$	-117.7 ± 0.1
Bayes factor $(aniso/iso)$	0.3

Table 3: Comparison of total rate inferences between anisotropic and isotropic models, together with log evidences and the resulting Bayes factor. Rates are quoted as medians with 90% credible intervals; log evidences are quoted with 1σ uncertainties.

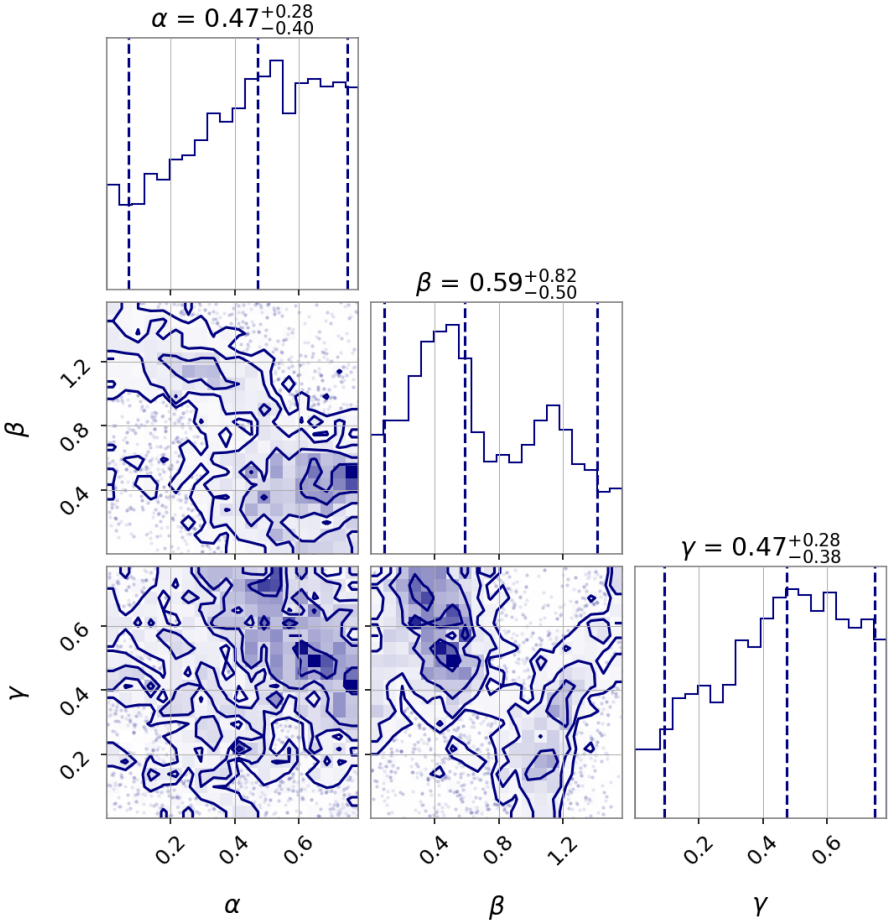


Figure 10: Above figure shows the 3 2D marginal distributions for each pair of Euler angles and the 3 1D marginal distributions for each of the Euler angles separately.

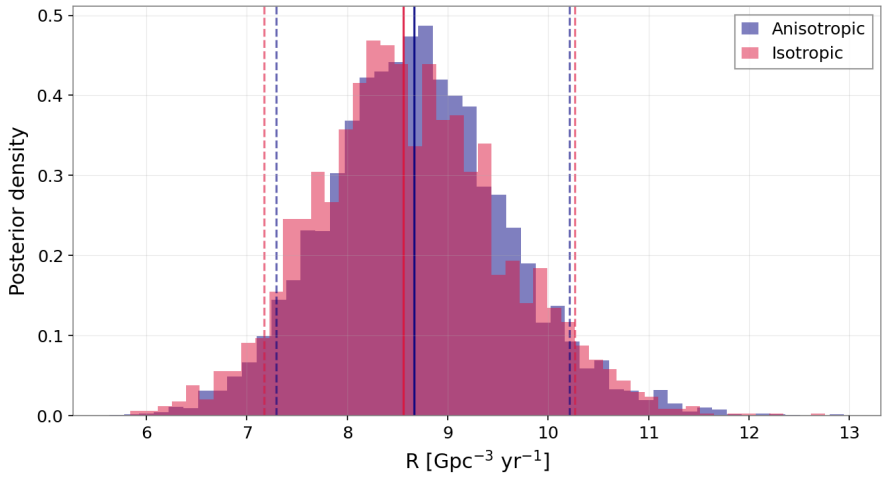


Figure 11: Posterior distribution histograms for the total rate R in both the isotropic and anisotropic model. Solid lines indicate the medians and the dashed lines indicate the 5th and 95th percentiles.

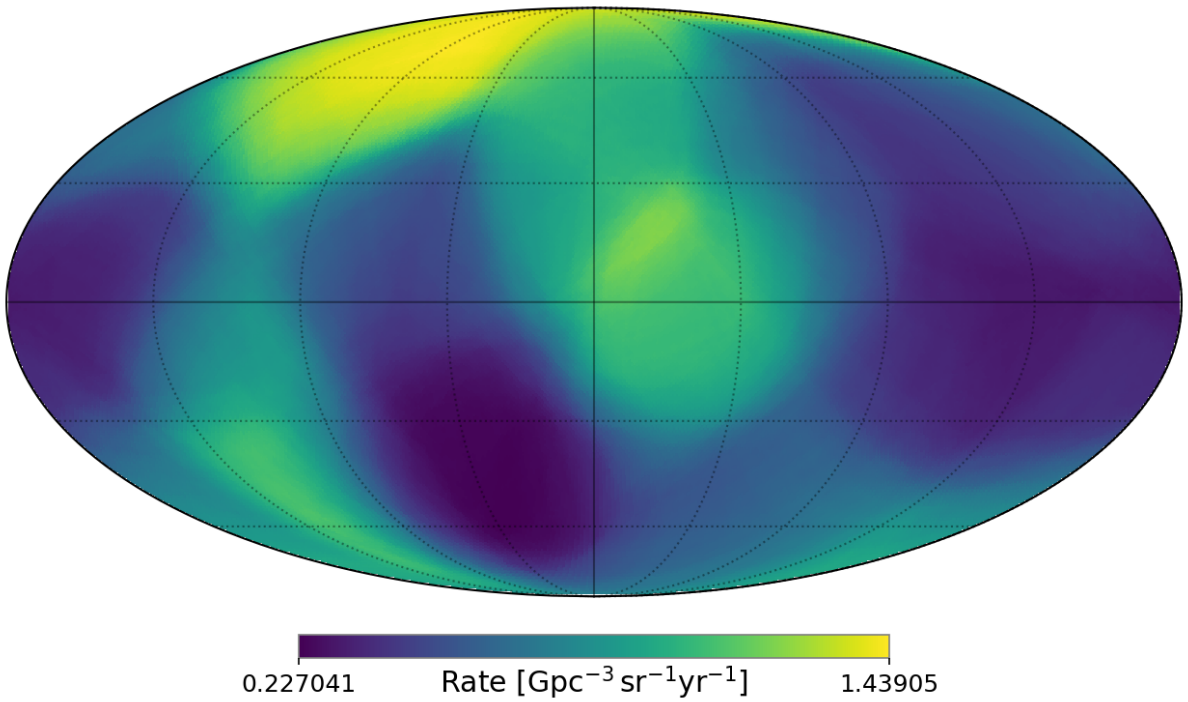
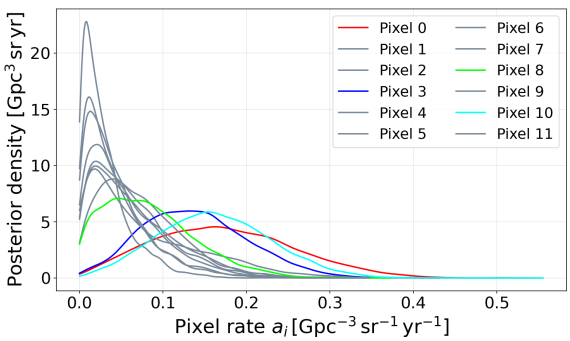
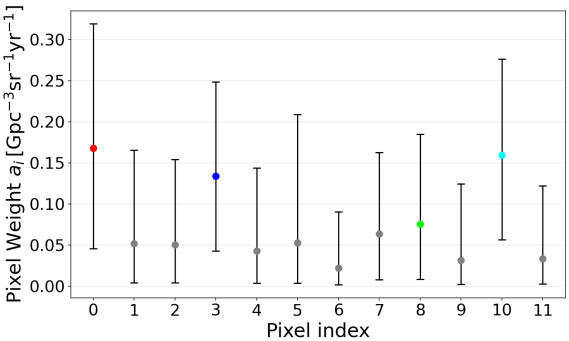


Figure 12: The above figure shows the differential rate across the sky averaged over the posterior samples for the pixel weights and the Euler angles.



(a) Posterior marginal distributions for each of the 12 pixel weights. The distributions of Pixel 0, 3, 8 and 10 are in colour.



(b) Median of the posteriors for each of the 12 pixel weights along with 5th to 95th credible intervals for each pixel.

Figure 13: Above plots represent the marginal posterior distributions for each of the pixel weights. Since we are neglecting selection effects the weights are bounded between 0 and 1.

4.2 No-Selection Effects Results

As a check, we decided to do the same analysis, without including any sort of selection effects into the likelihood function. In particular, without selection effects, the likelihood function doesn't include any probability of detection terms. So, we no longer include the number likelihood or the denominators of the event likelihood. The only parts that survive are the numerators of the individual event likelihoods shown in Equation 11.

In this simplified model, the likelihood only depends on the relative sizes of the pixel weights as opposed to their absolute sizes. We then restrict the pixel weights so that they sum to 1 and choose a Dirichlet$(1, 1, ..., 1)$ [8] prior such that they are distributed uniformly on the simplex $\sum_{i=1}^{N_{pix}} a_i = 1$. This means that the pixel weights provide 11 free parameters as opposed to the 15 we had for the full model. The Euler angles are included in the same way as in the full model.

$\log Z_{aniso}$	-205.2 ± 0.2
$\log Z_{iso}$	-206.2
Bayes factor $(aniso/iso)$	2.8

Table 4: Log evidences and the resulting Bayes factor for the analysis without selection effects included.

As shown in Table 4, for this simplified model, we compute a Bayes factor of 2.8 in favour of the anisotropic model over the isotropic model. Without consideration of selection effects, the data gives very weak preference for anisotropy. This is the opposite of what we find with the full likelihood.

Looking at Figure 4, these results are exactly what we would expect. As mentioned before, we can see a bias towards posterior samples in the northern and southern pixels. This is explained by Figure 6, which shows that the detectors observe a larger proportion of events originating from this part of the sky. However, in this new model, the differences across the sky shown in Figure 6 are not taken into account. The only data that is used in this inference are the posterior samples, which have a slightly uneven distribution across the sky. This is shown in Figure 4. It therefore makes perfect sense that the anisotropic model would be slightly preferred over isotropy.

As in Figure 9, we see in Figure 13 that Pixels 0, 3 and 10 have the largest values for their weights. The weight for Pixel 10 is almost equal to that of Pixel 0, which is different from the full inference. Additionally, the weight for Pixel 8 relative to the 8 smallest weights has decreased compared to in the full model. So, whilst the general shapes of the marginal posteriors are similar to those for the full model, there are clear differences in some of the relative sizes of the weights.

With no selection effects, the general shape of the marginal posteriors on the angles is unchanged with probability spread across the prior range. This can be seen in Appendix B.

4.3 Spherical Harmonics Model Results

As an additional experiment, we decided to abandon the discrete pixel weight model for describing the rates of GW events. Without considering selection effects, we are only interested in the probability of observing an event in some section of the sky as opposed the absolute values of the rates. We therefore want to consider a probability density function over the sphere describing how likely events are to originate from certain sky positions. Concretely, the probability of a GW source having sky position within some solid angle $d\Omega$ centred at (θ, ϕ) is $P(\theta, \phi)\, d\Omega$.

We write the log probability density, $\log(P(\theta, \phi))$, as a linear combination of the 3 dipole $(l = 1)$ spherical harmonics and the 5 quadrupole $(l = 2)$ spherical harmonics and the 7 spherical

harmonics of order 3 ($l = 3$):

$$\log(P(\theta, \phi)) = \sum_{l=1}^{3} \sum_{m=-l}^{m=l} a_{lm} Y_l^m(\theta, \phi) \tag{26}$$

So, now, our anisotropic model population parameters are these 15 coefficients for each of the 15 spherical harmonics we have in our basis. Of course, most choices of parameters do not yield normalised probability density functions. We therefore need to scale the resulting function to obtain a valid probability density. The map from coefficients to PDFs is therefore not injective.

In the two previous models, the distribution of GW sources is described discontinuously using a `HEALPix` grid. Effectively, we decompose the density function as a linear combinations of indicator functions on the sphere. On the other hand, spherical harmonics are smooth and therefore yield continuous probability distributions. This means that implementation of the code is much more straightforward. Additionally, this approach may be preferable since it allows direct comparison with the power spectrum of the CMB anisotropy results.

In order to compute the likelihood, we do the same thing as we did for the no-selection model. For each event we simply compute the Monte Carlo sum in Equation 11. Instead of working out which pixel contains each of the samples and then summing the pixel rates (as in Equation 12), we evaluate all 15 of the spherical harmonics at each of our posterior samples. Evaluating Equation 11 for a particular set of coefficients is then a simple matrix calculation.

Figure 14 suggests that there is a preference for non-zero anisotropy, dominated by large positive values for $a_{2,0}$. The density over the sky, averaged over our different posterior samples for the coefficients, exhibits an unusually large peak near the celestial north pole, with reduced but fairly uniform density elsewhere.

To assess whether the polar enhancement is a feature of the basis rather than a global maximum found by the sampler, we considered the empirical density given directly by the HEALPix histogram of posterior samples from the 83 events used in our analysis (shown in Figure 15). This choice of probability density yields a high event likelihood, since it places probability mass exactly where the samples land. So, in an attempt to approximate this empirical density function using a linear combination of spherical harmonics, we projected it onto the $L = 3$ basis (via least-squares fitting in log-density space). The resulting density function across the sky yields a substantially lower likelihood than the true empirical distribution. The maximum likelihood sample given by the nested sampler also has a higher likelihood than this projection approximation but is still well below the likelihood of the true empirical distribution in Figure 15. This confirms that the basis itself, rather than the sampler, is the limiting factor. The $L = 3$ expansion lacks the angular resolution to fully represent the catalogue's spatial structure. As noted in Payne et al [9], higher order decompositions ($L = 5$ and beyond) would be able to capture finer features.

Despite the obvious limitations of the current model, Table 5 shows a much more significant preference for the anisotropic model when we use a basis of spherical harmonics up to $L = 3$. These results match those we obtained in our other analysis without consideration of selection effects. This is consistent with our prediction that, without selection effects, the anisotropic model would be preferred.

$\log(Z_{aniso})$	-208.2
$\log(Z_{iso})$	-210.1
Bayes factor ($aniso/iso$)	6.9

Table 5: Table above shows the evidences for each model and Bayes factor for our spherical harmonics analysis.

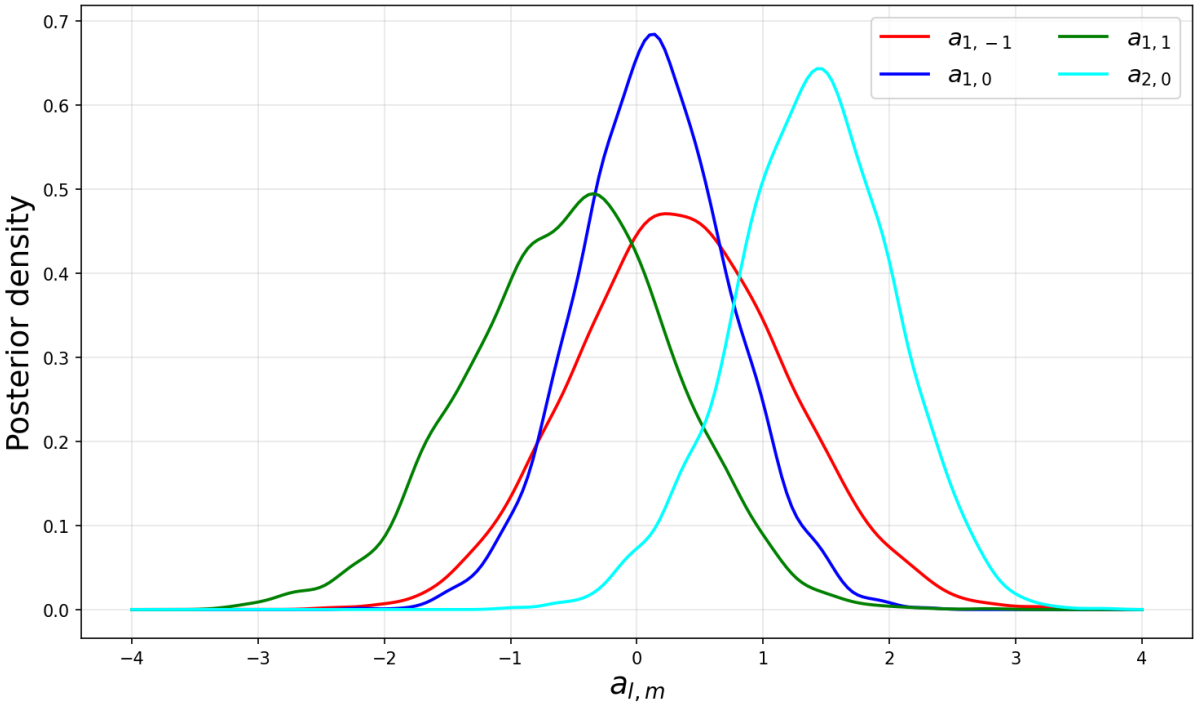


Figure 14: Marginal Posteriors for 4 of the spherical harmonics coefficients, $a_{1,-1}$, $a_{1,0}$, $a_{1,1}$, $a_{2,0}$. Of the 15 spherical harmonic coefficients considered in the analysis, we only plot the marginals for the 3 dipole coefficients and $a_{2,0}$ whose marginal posterior is centred furthest from 0.

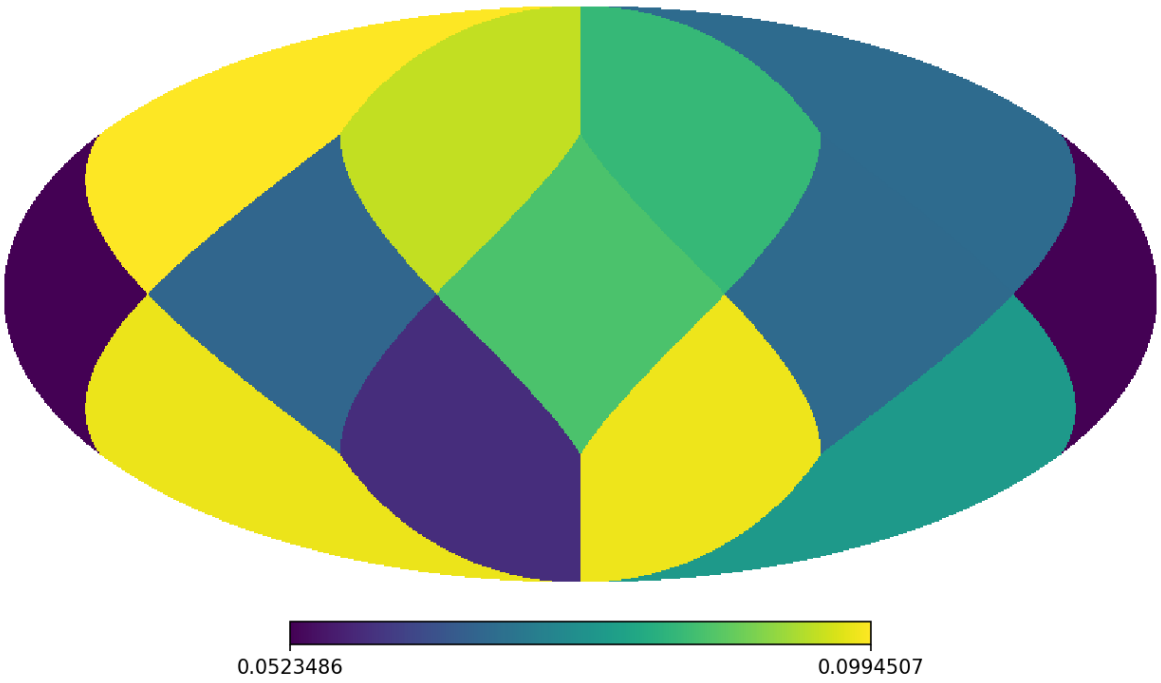


Figure 15: Mollweide projection showing the fractions of posterior samples from the 83 events used in our analysis in each pixel. We refer to this as the 'empirical distribution' below.

5 Conclusion

We performed 3 separate Bayesian model selections. Our main result is the Bayes factor of 0.3 (Z_{aniso}/Z_{iso}) we found for the full likelihood function, which includes selection effects. This suggests a slight preference for isotropy over anisotropy but is inconclusive and gives no significant evidence for one model over the other. As mentioned previously, this is consistent with previous research (e.g. R. Stiskalek et al [1], Z. Li et al [2], K. Gopal et al [3], E. Payne et al [9]). These results were expected due to the exceptional uniformity of the CMB and also the past research using GWs.

Other than the main results including selection effects, we performed two other separate analyses without consideration of selection effects. As a result, these results are interesting but much less significant. We find two Bayes factors of 2.8 and 6.9 in favour of anisotropy for the `healpy` pixel model and the spherical harmonics model respectively. Whilst, as expected, these no selection models give slight preferences for anisotropy, there is still no significant evidence for either model. These side results further reinforce our main conclusion.

The increasing sensitivity of the detectors resulting in higher rates of GW detection will result in larger and larger catalogues of GW events in single observing runs. This increase in data will lead to more reliable inferences on isotropy of these sources without any additional complications. On the other hand, a way of incorporating more data currently available into the analysis would be to compute detection probabilities across different observing runs that are completely consistent with the different numbers of events, durations and detector sensitivities between runs. This is unlikely to change the conclusions of the analysis. However, it is essential if one wishes to use all of the data at your disposal.

Another way of improving the analysis is to use a different PSD for each of the different events in the catalogue. This is the correct approach for computing the denominator of the event likelihood in Equation 10 and the detection probabilities in Equation 20. As noted previously, we make an approximation by using a fixed PSD, which was calculated using strain data from the O4 run. In theory, one can extract PSDs for each event from the hdf5 files and calculate the detection probability grid using 83 distinct PSDs. However, their availability across events is inconsistent, which makes it technically cumbersome to write code that cleanly obtains the right PSDs for each of the active detectors. This approximation may introduce a small bias to the size of the inferred rates but is unlikely to affect conclusions regarding anisotropy. Using separate PSDs also increases the run time of the likelihood function. To compensate, we may have been forced to reduce the number of posterior samples used from each event, which increases the variance of the results.

We conclude that there is no clear evidence for either an isotropic or an anisotropic model. Whilst this is consistent with the large majority of previous research, the future availability of more data as well as improved localisation of sources means future population inferences will become more reliable. Consequently, we may find that repeating this analysis in the future yields different conclusions.

Acknowledgements

During the development of this project, ChatGPT and Claude were used to assist in debugging and optimising certain sections of the code used for our analysis.

A Duplication of GW230605 Sanity Check

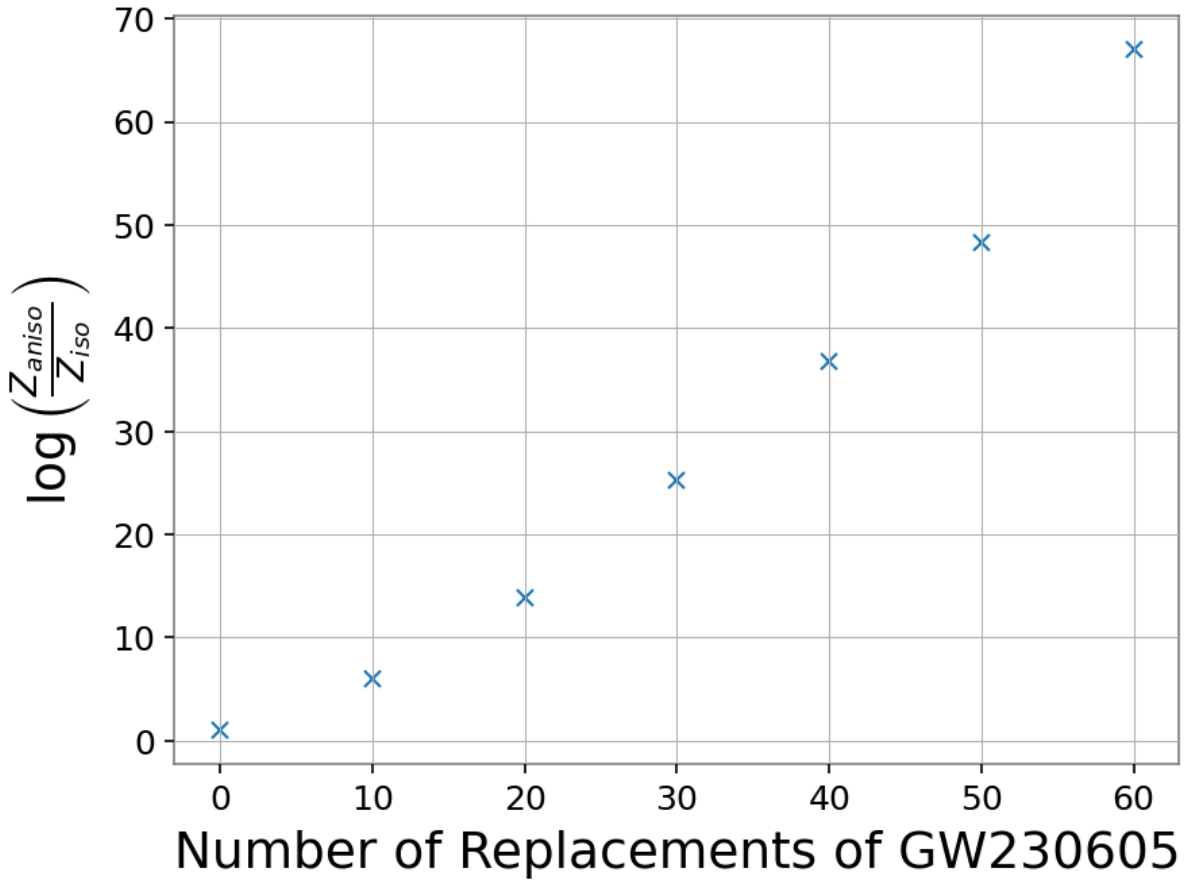


Figure 16: Logarithm of the Bayes factor plotted against the number of times we duplicate the event GW230605 in our catalogue.

For different values of n, we replace the first n events in our full catalogue with n copies of the first event in our catalogue, GW230605. With this edited dataset, we compute the Bayes factor for the no selection pixelated model. As expected, Figure 16 shows that as we increase the number of duplications, the Bayes factor (Z_{aniso}/Z_{iso}) increases exponentially. As the number of copies of GW230605 increases, the full set of posterior samples becomes more concentrated in the specific area of the sky corresponding to GW230605. The distribution of posterior samples across the sky then appears much more anisotropic. As a result, the anisotropic model is increasingly preferred by the data.

B Full Likelihood Corner Plot

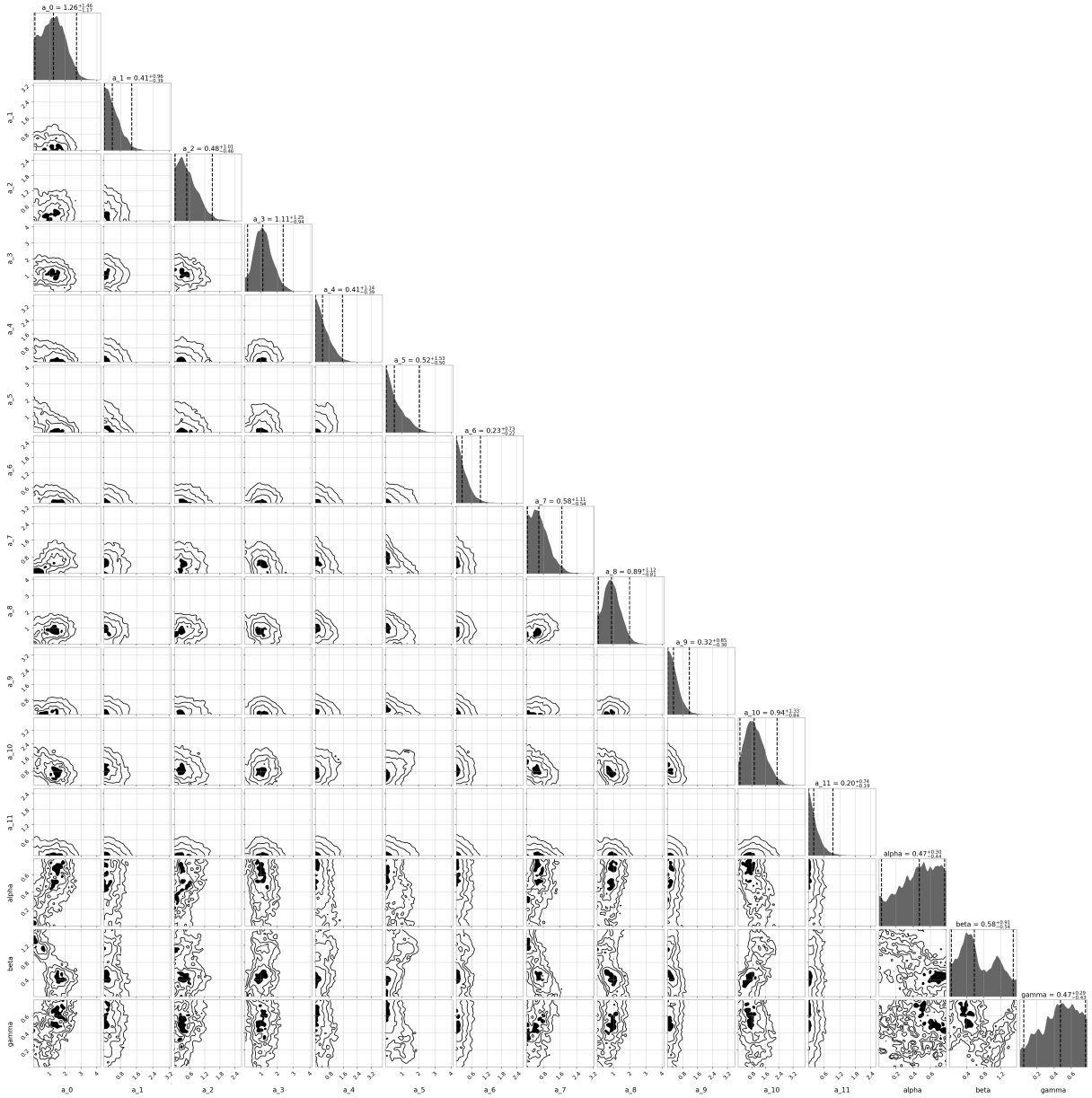


Figure 17: Joint posterior on the pixel rates and Euler angles under the anisotropic model. These are results for the full model.

C No Selection Effects Corner Plot

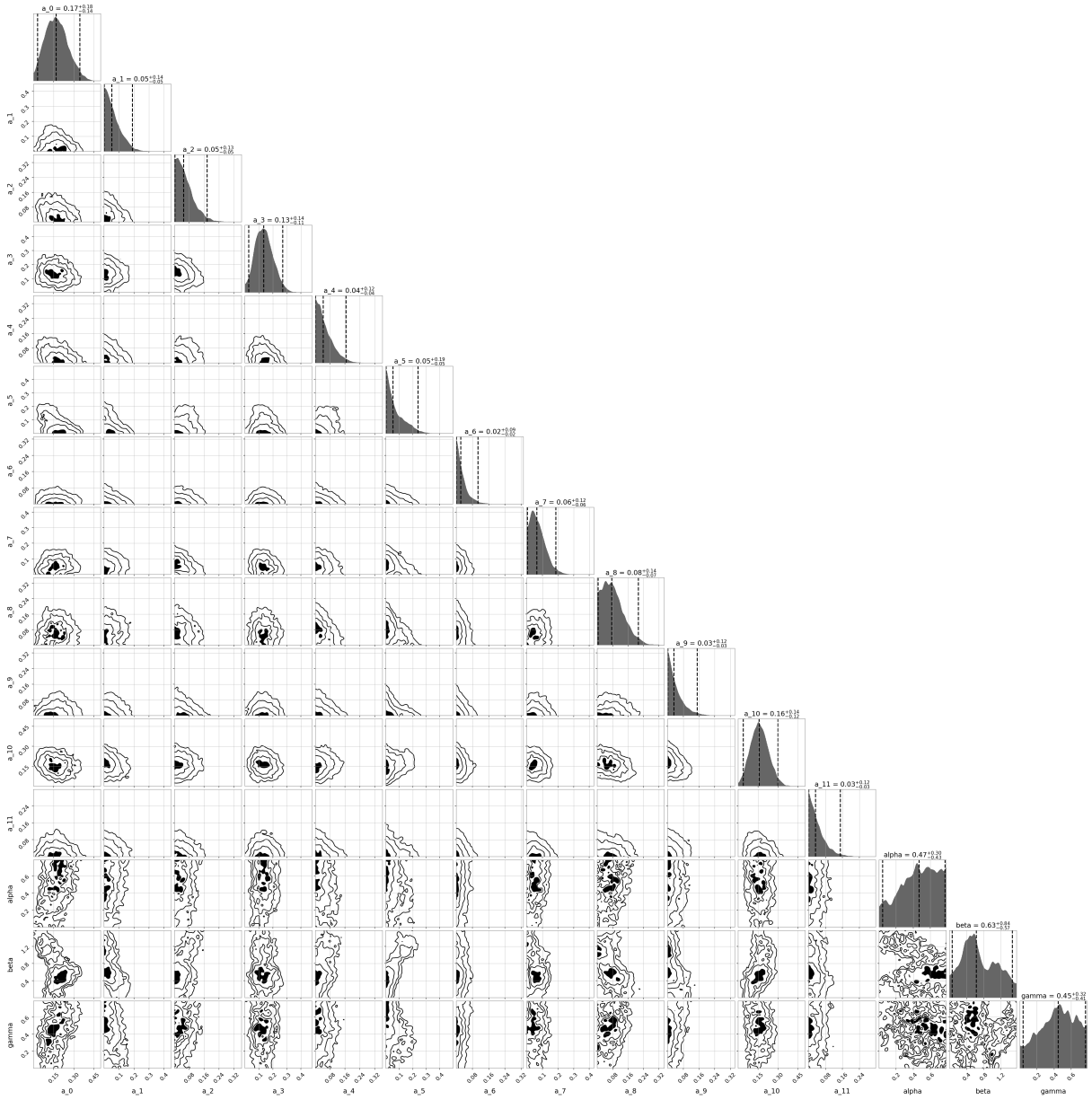


Figure 18: Joint posteriors on the pixel weights and Euler angles in the anisotropic model without selection effects.

D Spherical Harmonics Corner Plot

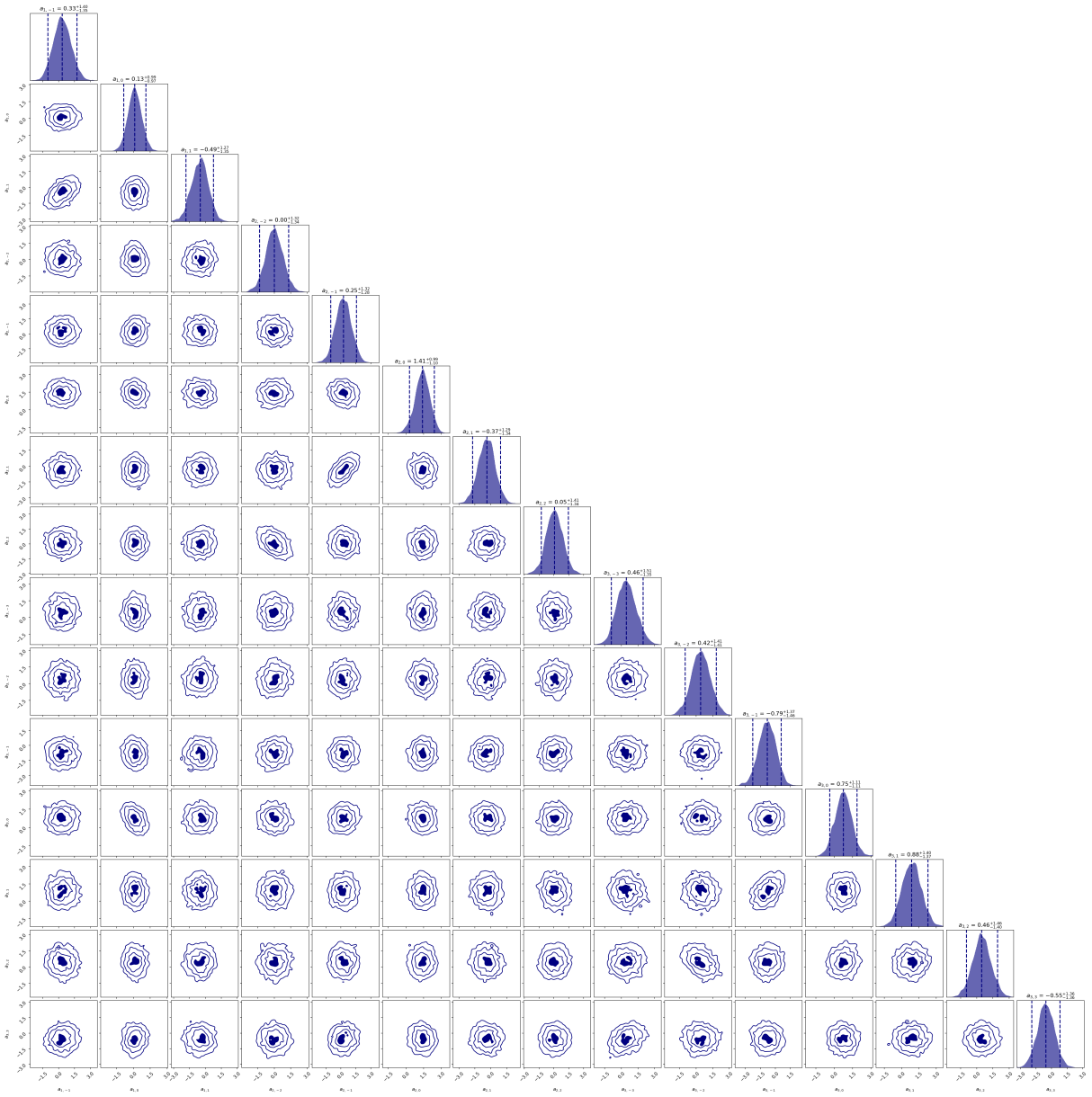


Figure 19: Joint posteriors on the coefficients of each of the spherical harmonics. This is for the model that uses a truncated basis of spherical harmonics to describe the spatial distribution of GW sources without consideration of selection effects (Section 4.3).

E Table of Events

EVENT NAME	ACTIVE DETECTORS
GW230605_065343	HL
GW230606_004305	HL
GW230608_205047	HL
GW230609_064958	HL
GW230624_113103	HL
GW230627_015337	HL
GW230628_231200	HL
GW230630_070659	HL
GW230630_125806	HL
GW230630_234532	HL
GW230702_185453	HL
GW230704_021211	HL
GW230704_212616	HL
GW230706_104333	HL
GW230707_124047	HL
GW230708_053705	HL
GW230708_230935	HL
GW230709_122727	HL
GW230712_090405	HL
GW230723_101834	HL
GW230726_002940	L
GW230729_082317	HL
GW230731_215307	HL
GW230803_033412	HL
GW230805_034249	HL
GW230806_204041	HL
GW230811_032116	HL
GW230814_061920	L
GW230814_230901	HL
GW230819_171910	HL
GW230820_212515	HL
GW230824_033047	HL
GW230825_041334	HL
GW230831_015414	HL
GW230904_051013	HL
GW230911_195324	HL
GW230914_111401	HL
GW230919_215712	HL

Table 6: List of the first half of the events for which posterior sample data was used in our analysis. If H is present in the right hand column then the Hanford detector was active. If L is present then the Livingston detector was active.

EVENT NAME (continued)	ACTIVE DETECTORS (continued)
GW230920_071124	HL
GW230922_020344	HL
GW230922_040658	HL
GW230924_124453	HL
GW230927_043729	HL
GW230927_153832	HL
GW230928_215827	HL
GW230930_110730	HL
GW231001_140220	HL
GW231004_232346	HL
GW231005_021030	HL
GW231005_091549	HL
GW231008_142521	HL
GW231014_040532	HL
GW231018_233037	HL
GW231020_142947	HL
GW231028_153006	HL
GW231029_111508	L
GW231102_071736	HL
GW231104_133418	HL
GW231108_125142	HL
GW231110_040320	HL
GW231113_122623	HL
GW231113_200417	HL
GW231114_043211	HL
GW231118_005626	HL
GW231118_071402	HL
GW231118_090602	HL
GW231119_075248	HL
GW231123_135430	HL
GW231127_165300	HL
GW231129_081745	HL
GW231206_233134	HL
GW231206_233901	HL
GW231213_111417	HL
GW231221_135041	HL
GW231223_032836	HL
GW231223_075055	HL
GW231223_202619	H
GW231224_024321	HL
GW231226_101520	HL
GW231230_170116	HL
GW231231_154016	H
GW240104_164932	H
GW240107_013215	HL
GW240109_050431	H

Table 7: List of the second half of the events for which posterior sample data was used in our analysis. If H is present in the right hand column then the Hanford detector was active. If L is present then the Livingston detector was active.

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